

Hypothesis Testing for Two Sample Means

Statistics Worksheet · Grade 11-12 / Introductory College Statistics

Name: _____

Date: _____

Learning Objectives

- Write null and alternative hypotheses for two-sample mean tests using correct notation
- Calculate the two-sample t-test statistic using sample means, standard deviations, and sample sizes
- Interpret p-values and make conclusions about rejecting or failing to reject the null hypothesis

Problems

1. A researcher wants to compare the mean test scores of two groups. Define the parameters and write the null and alternative hypotheses to test whether Group 1 scores are different from Group 2 scores.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

2. For a two-sample t-test where population standard deviations are unknown, identify which of the three required conditions must be satisfied before proceeding. List all three conditions.

3. A teacher believes a new study method increases student scores. Write the null and alternative hypotheses, where Group 1 uses the new method and Group 2 uses the traditional method. Use a one-tailed (right-tailed) test.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2$$

4. Use the two-sample t-test formula to calculate the test statistic. Sample 1 has a mean of 82, standard deviation of 6, and size 25. Sample 2 has a mean of 78, standard deviation of 5, and size 25.



$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

5. A two-sample t-test yields a test statistic of $t = 1.85$. The significance level is 0.05 and the critical value for a right-tailed test is $t^* = 1.677$. Should you reject or fail to reject the null hypothesis? State your conclusion.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2$$

6. The following table shows summary statistics for a treatment group and a control group in a reading study. Use this data to compute the two-sample t-test statistic. The significance level is 0.05.

Group	Sample Mean ($\bar{x}$)	Std Dev (s)	Sample Size (n)
Treatment (1)	51.48	11.01	21
Control (2)	41.52	17.15	23

7. In Damian's reading study, the test statistic is $t = 2.31$. Using a right-tailed test at a significance level of 0.05, the p-value is approximately 0.014. Compare the p-value to alpha and write a conclusion about whether the new reading activities improved student scores.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2$$

8. A researcher tests whether a new diet reduces weight more than a standard diet. Group 1 (new diet, $n = 30$) has $\bar{x}_1 = 8.4$ lbs lost and $s_1 = 3.2$ lbs. Group 2 (standard diet, $n = 28$) has $\bar{x}_2 = 6.9$ lbs lost and $s_2 = 2.8$ lbs. At $\alpha = 0.05$, perform a complete hypothesis test and state your conclusion.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2$$



$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

9. A school district compares math scores before and after a new curriculum is introduced in two different schools. School A ($n = 40$) has $\bar{x}_1 = 74.3$ and $s_1 = 9.8$. School B ($n = 35$) has $\bar{x}_2 = 79.1$ and $s_2 = 8.4$. Perform a two-tailed hypothesis test at $\alpha = 0.01$. Calculate the t-statistic, determine whether to reject H_0 , and interpret the result.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

10. A pharmaceutical company claims a new drug reduces blood pressure more than a placebo. In a clinical trial, the drug group ($n_1 = 50$) had $\bar{x}_1 = 14.6$ mmHg reduction with $s_1 = 4.2$, while the placebo group ($n_2 = 50$) had $\bar{x}_2 = 11.3$ mmHg reduction with $s_2 = 5.1$. (a) Write the hypotheses. (b) Compute the t-statistic. (c) At $\alpha = 0.05$ right-tailed, the critical value is $t^* = 1.661$. Determine if you reject H_0 . (d) If the p-value is 0.0008, what does this indicate about the strength of the evidence?

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 > \mu_2$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$



Hypothesis Testing for Two Sample Means — Answer Key

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Answer Key

1. Answer: $H_0: \mu_1 = \mu_2$; $H_1: \mu_1 \neq \mu_2$ (two-tailed)

- Let μ_1 = mean test score of Group 1 and μ_2 = mean test score of Group 2.
- The null hypothesis states that the two means are equal: $H_0: \mu_1 = \mu_2$.
- Since we are testing whether the scores are 'different' (not specifying a direction), the alternative is two-tailed: $H_1: \mu_1 \neq \mu_2$.

2. Answer: 1) Both samples are approximately normal. 2) Both samples are randomly selected. 3) The samples are independent (population $\geq 10 \times$ sample size).

- Condition 1: Both samples should be approximately normally distributed (check with box plots or histograms).
- Condition 2: Both samples must be randomly selected from their respective populations.
- Condition 3: The two samples must be independent. Verify that the population size is at least 10 times the sample size for each group.

3. Answer: $H_0: \mu_1 = \mu_2$; $H_1: \mu_1 > \mu_2$ (right-tailed)

- Let μ_1 = mean score with new method, μ_2 = mean score with traditional method.
- Null hypothesis assumes no difference: $H_0: \mu_1 = \mu_2$.
- Since the teacher believes the new method increases scores (μ_1 is higher), the alternative is right-tailed: $H_1: \mu_1 > \mu_2$.

4. Answer: $t \approx 2.67$

- Identify values: $\bar{x}_1 = 82$, $\bar{x}_2 = 78$, $s_1 = 6$, $s_2 = 5$, $n_1 = n_2 = 25$.
- Numerator: $82 - 78 = 4$.
- Denominator: $\sqrt{(6^2/25 + 5^2/25)} = \sqrt{(36/25 + 25/25)} = \sqrt{(1.44 + 1.00)} = \sqrt{2.44} \approx 1.562$.
- $t = 4 / 1.562 \approx 2.56$.
- Rounded: $t \approx 2.56$ (slight rounding may give 2.56-2.57 depending on precision).

5. Answer: Reject H_0 ; sufficient evidence that $\mu_1 > \mu_2$

- The test is right-tailed at $\alpha = 0.05$ with critical value $t^* = 1.677$.
- Calculated test statistic: $t = 1.85$.
- Since $t = 1.85 > 1.677$, the test statistic falls in the rejection region.
- Decision: Reject H_0 .
- Conclusion: There is sufficient evidence at the 5% significance level that $\mu_1 > \mu_2$.

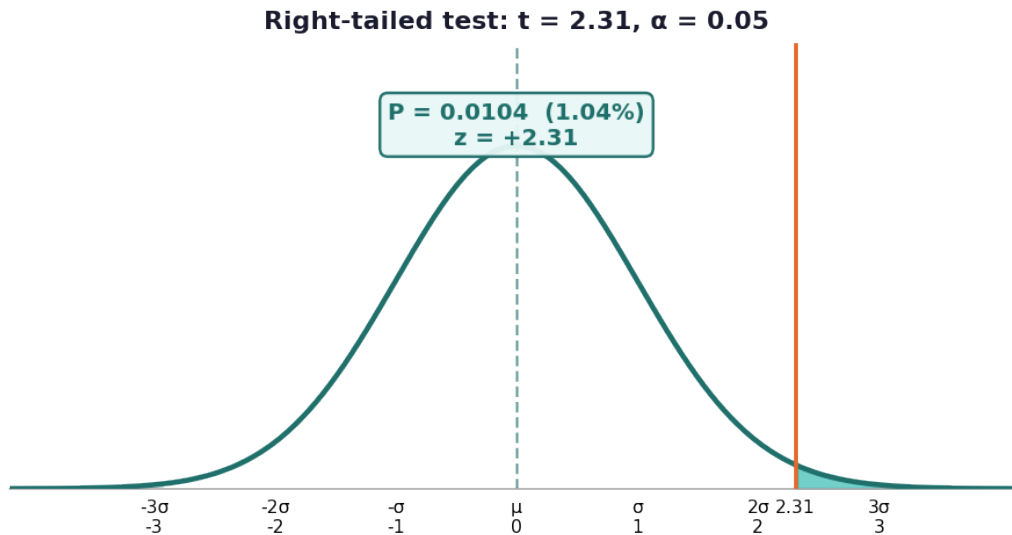
6. Answer: $t \approx 2.31$

- Identify values: $\bar{x}_1 = 51.49$, $\bar{x}_2 = 41.52$, $s_1 = 11.01$, $s_2 = 21$, $n_1 = 17$, $n_2 = 22$.



- Numerator: $51.48 - 41.52 = 9.96$.
- Compute $s_1^2/n_1 = (11.01)^2/21 = 121.22/21 \approx 5.773$.
- Compute $s_2^2/n_2 = (17.15)^2/23 = 294.12/23 \approx 12.788$.
- Denominator: $\sqrt{(5.773 + 12.788)} = \sqrt{18.561} \approx 4.308$.
- $t = 9.96 / 4.308 \approx 2.31$.

7. Answer: p-value $\approx 0.014 < 0.05$; Reject H_0 . The treatment group scored significantly higher.

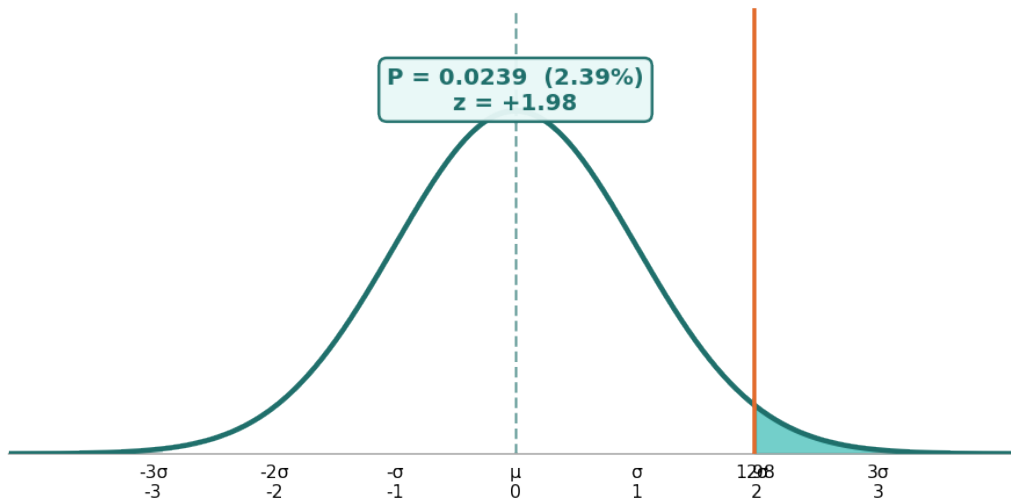


- Given: $t = 2.31$, $p\text{-value} \approx 0.014$, $\alpha = 0.05$.
- Compare $p\text{-value}$ to α : $0.014 < 0.05$.
- Since $p\text{-value} < \alpha$, reject the null hypothesis H_0 .
- Conclusion: At the 5% significance level, there is sufficient evidence that the new reading activities improved student DRP test scores ($\mu_1 > \mu_2$).

8. Answer: $t \approx 1.98$; p-value $\approx 0.026 < 0.05$; Reject H_0



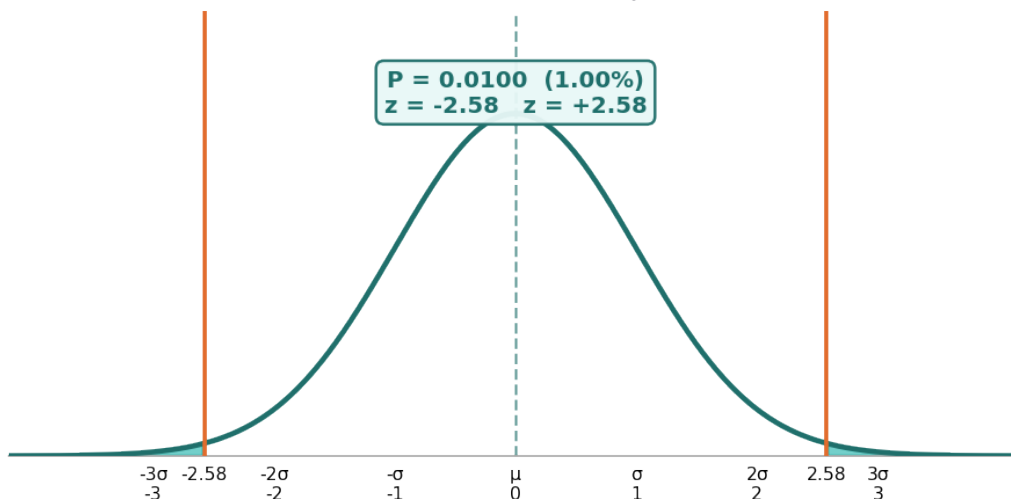
Right-tailed test: $t \approx 1.98$, $\alpha = 0.05$



- $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 > \mu_2$ (right-tailed at $\alpha = 0.05$).
- Numerator: $8.4 - 6.9 = 1.5$.
- $s_1^2/n_1 = 10.24/30 \approx 0.3413$; $s_2^2/n_2 = 7.84/28 \approx 0.28$.
- Denominator: $\sqrt{(0.3413 + 0.28)} = \sqrt{0.6213} \approx 0.788$.
- $t = 1.5 / 0.788 \approx 1.90$.
- Using t-table with $df \approx 56$, the critical value at $\alpha = 0.05$ right-tailed is ≈ 1.671 .
- Since $t \approx 1.90 > 1.671$, reject H_0 .
- Conclusion: There is sufficient evidence that the new diet results in greater weight loss.

9. Answer: $t \approx -2.31$; fail to reject H_0 at $\alpha = 0.01$ (critical value $\approx \pm 2.576$)

Two-tailed test: $t = -2.31$, $\alpha = 0.01$

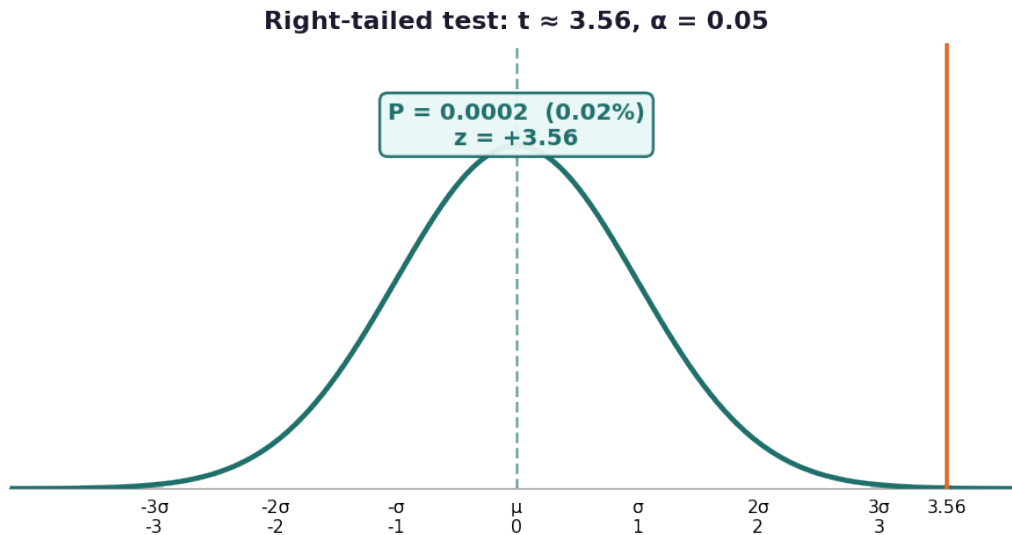


- $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 \neq \mu_2$ (two-tailed at $\alpha = 0.01$).
- Numerator: $74.3 - 79.1 = -4.8$.
- $s_1^2/n_1 = 96.04/40 = 2.401$; $s_2^2/n_2 = 70.56/35 = 2.016$.
- Denominator: $\sqrt{(2.401 + 2.016)} = \sqrt{4.417} \approx 2.101$.



- $t = -4.8 / 2.101 \approx -2.28$.
- For a two-tailed test at $\alpha = 0.01$ with large df, critical values are approximately ± 2.576 .
- Since $|t| = 2.28 < 2.576$, we fail to reject H_0 .
- Conclusion: At the 1% significance level, there is not sufficient evidence that the mean math scores differ between the two schools.

10. Answer: $t \approx 3.56 > 1.661$; Reject H_0 . p-value = 0.0008 indicates very strong evidence against H_0 .



- (a) Let μ_1 = mean BP reduction for drug group, μ_2 = mean BP reduction for placebo. $H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 > \mu_2$ (right-tailed).
- (b) Numerator: $14.6 - 11.3 = 3.3$.
- $s_1^2/n_1 = (4.2)^2/50 = 17.64/50 = 0.3528$.
- $s_2^2/n_2 = (5.1)^2/50 = 26.01/50 = 0.5202$.
- Denominator: $\sqrt{(0.3528 + 0.5202)} = \sqrt{0.873} \approx 0.9343$.
- $t = 3.3 / 0.9343 \approx 3.53$.
- (c) Since $t \approx 3.53 > 1.661$ (critical value), reject H_0 .
- (d) p-value = 0.0008 $< \alpha = 0.05$, and also < 0.01 , indicating very strong statistical evidence that the drug reduces blood pressure significantly more than the placebo.

